Axis 1 holds : Second belt assembly , 2 motors , power screw assembly

tot load $\approx$ 2.07 Kg (assuming motor mass = 500 g)



$\Sigma T = J\alpha$; $T_m - T_L = (J_L + J_m)\alpha$

$J = mr^2 \Longrightarrow$ m = mass of load ; r = radius of pulley

$J_L = 2070 \times 0.637^2 = 840\ g\,cm^2$ ①

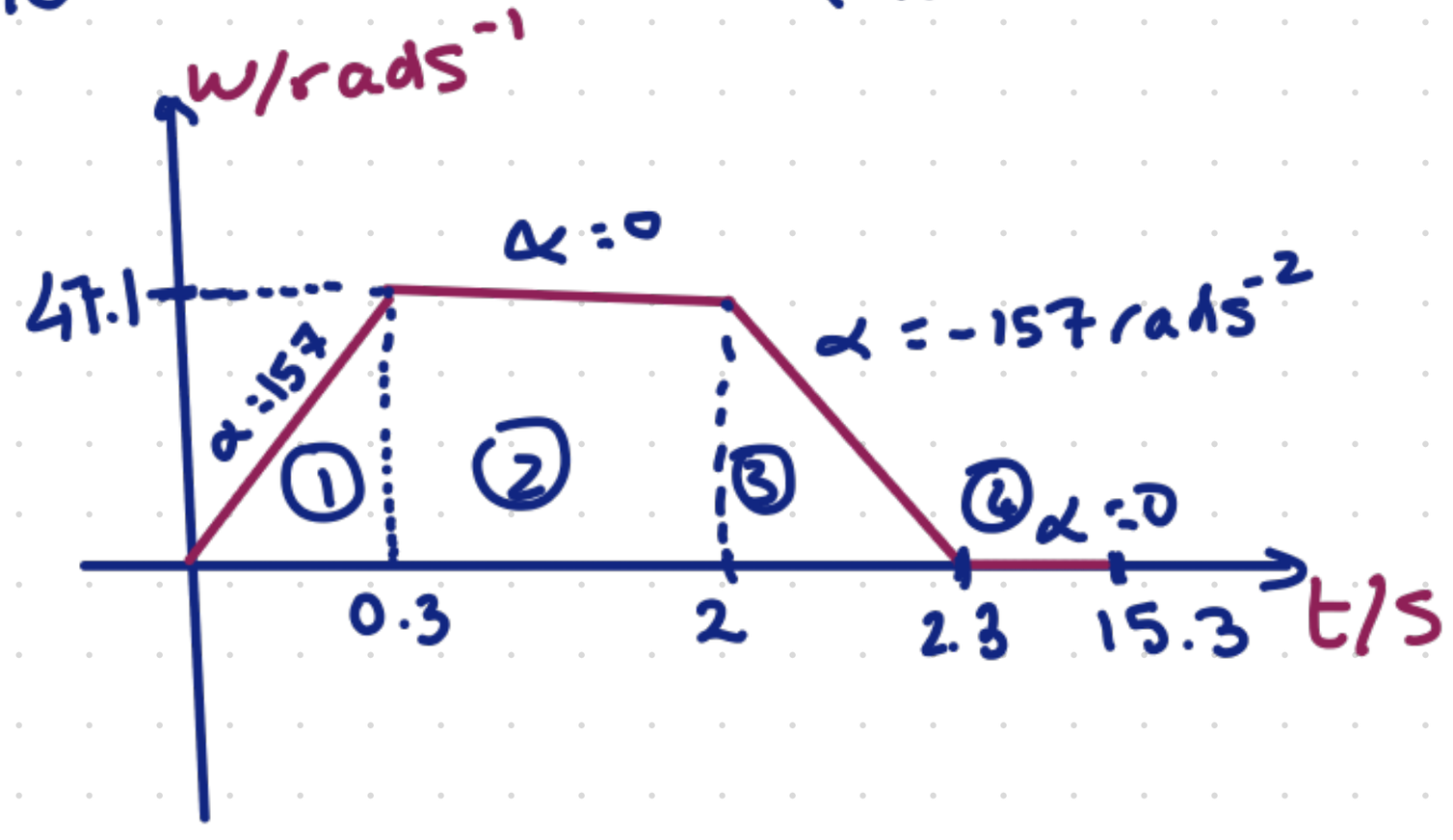
↓ radius of Std pulley

let $J_m = J_L = 840\ g\,cm^2$; $J_{tot} = 1680\ g\,cm^2$

$$T_L = mg\mu r = 2.07 \times 9.81 \times 0.05 \times 0.00637$$

$$T_L = 0.65 Ncm$$

Velocity Profile: (tot dist = 60 cm)



$$T_m = T_L + J_{tot}\,\alpha = 0.65\ N\text{cm} + 1680\times10^{-7}\times\alpha$$

$$T_1 = 0.65 + \left[1680\times10^{-7}\,(157)\right]\times100 = 3.2\ N\ \text{cm}$$

$$T_2 = 0.65\ N\ \text{cm}$$

$$T_3 = 0.65 + 1680\times10^{-5}\times(-157) = -1.9876\ N\text{cm}$$

$$T_4 = 0.65\ N\text{cm}$$

$$T_{rms} = \sqrt{\frac{3.2^2\times0.3 + 0.65^2\times1.7 + 1.9876^2\times0.3 + 0.65^2\times13}{6.4}}$$

$$T_{rms} = 1.28\ N\text{cm}$$

⇒ chosen actuator - NEMA17

Axis 2 holds : 1 motor , power screw assembly

tot load $\approx$ 1.0 kg (assuming motor mass = 500 g)



$\Sigma T = J\alpha$; $T_m - T_L = (J_L + J_m)\alpha$

$J_L = mr^2 \Longrightarrow$ m = mass of load ; r = radius of pulley

$J_L = 1000 \times 0.637^2 = 405.7\ g\,cm^2$ ①

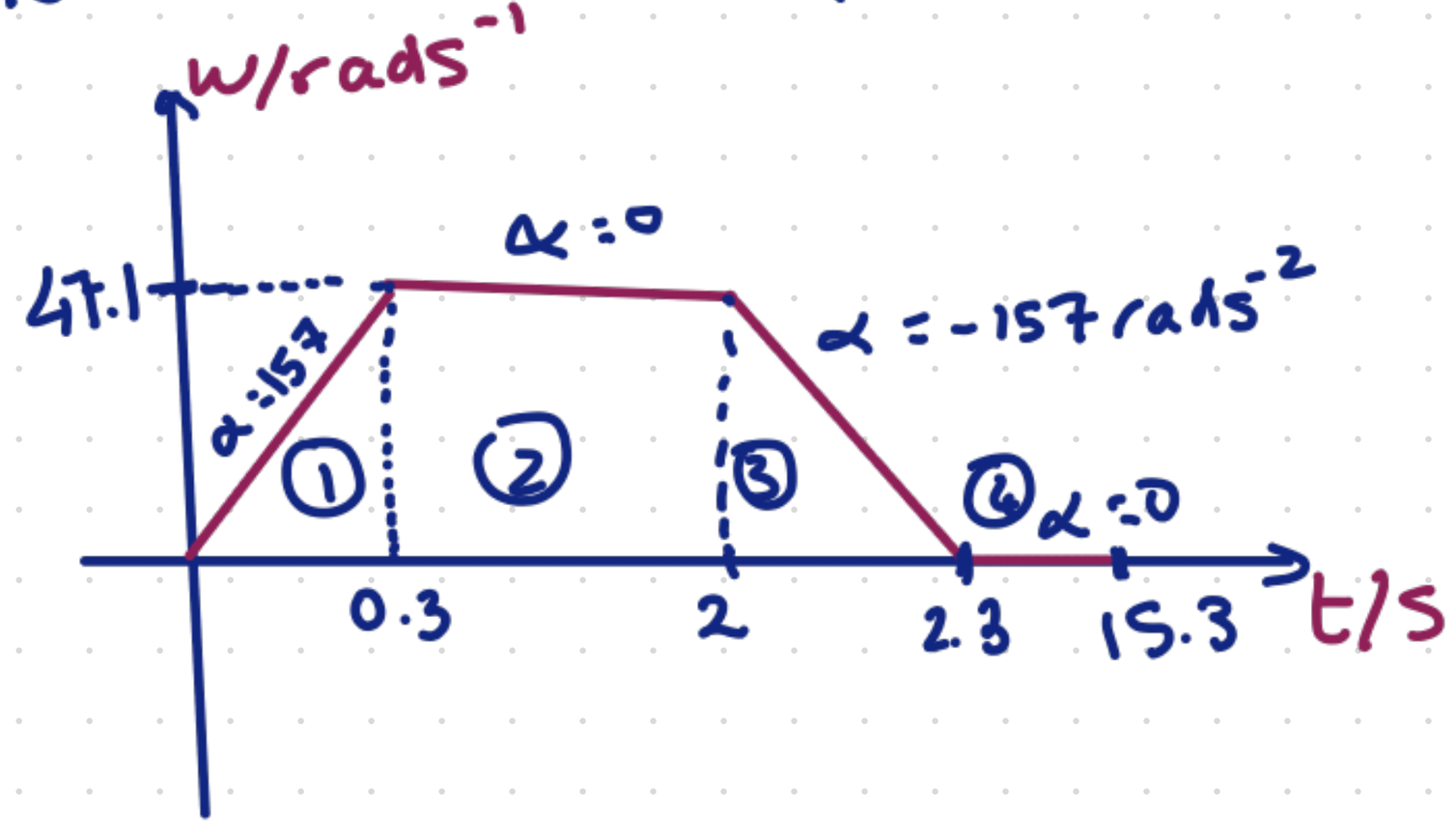
↓ radius of Std pulley

let $J_m = J_L = 405.7\ g\,cm^2$; $J_{tot} = 811.4\ g\,cm^2$

$$T_L = mg\mu r = 2 \times 9.81 \times 0.05 \times 0.00637$$

$$T_L = 0.31 \text{ Ncm}$$

Velocity Profile: (tot dist = 60 cm)



$$T_m = T_L + J_{tot}\,\alpha = 0.31\ Ncm + 811.4\times10^{-7}\times\alpha$$

$$T_1 = 0.31 + \left[811.4\times10^{-7}(157)\right]\times100 = 1.58\ N\,cm$$

$$T_2 = 0.31\ N\,cm$$

$$T_3 = 0.31 + 811.4\times10^{-5}\times(-157) = -0.97\ N\,cm$$

$$T_4 = 0.31\ N\,cm$$

$$T_{rms} = \sqrt{\frac{1.58^2\times0.3 + 0.31^2\times1.7 + 0.97^2\times0.3 + 0.31^2\times13}{6.4}}$$

$$T_{rms} = 0.62\ N\,cm$$

⇒ chosen actuator - NEMA17

NEMA 17 inertia = 54 g cm² (from datasheet)

which is less than both assumed inertias

Axis 3 holds : product, carriage, vaccum pump

mass = 0.4 kg



Power screw

$$T_m - T_L = (J_m + J_L)\alpha$$

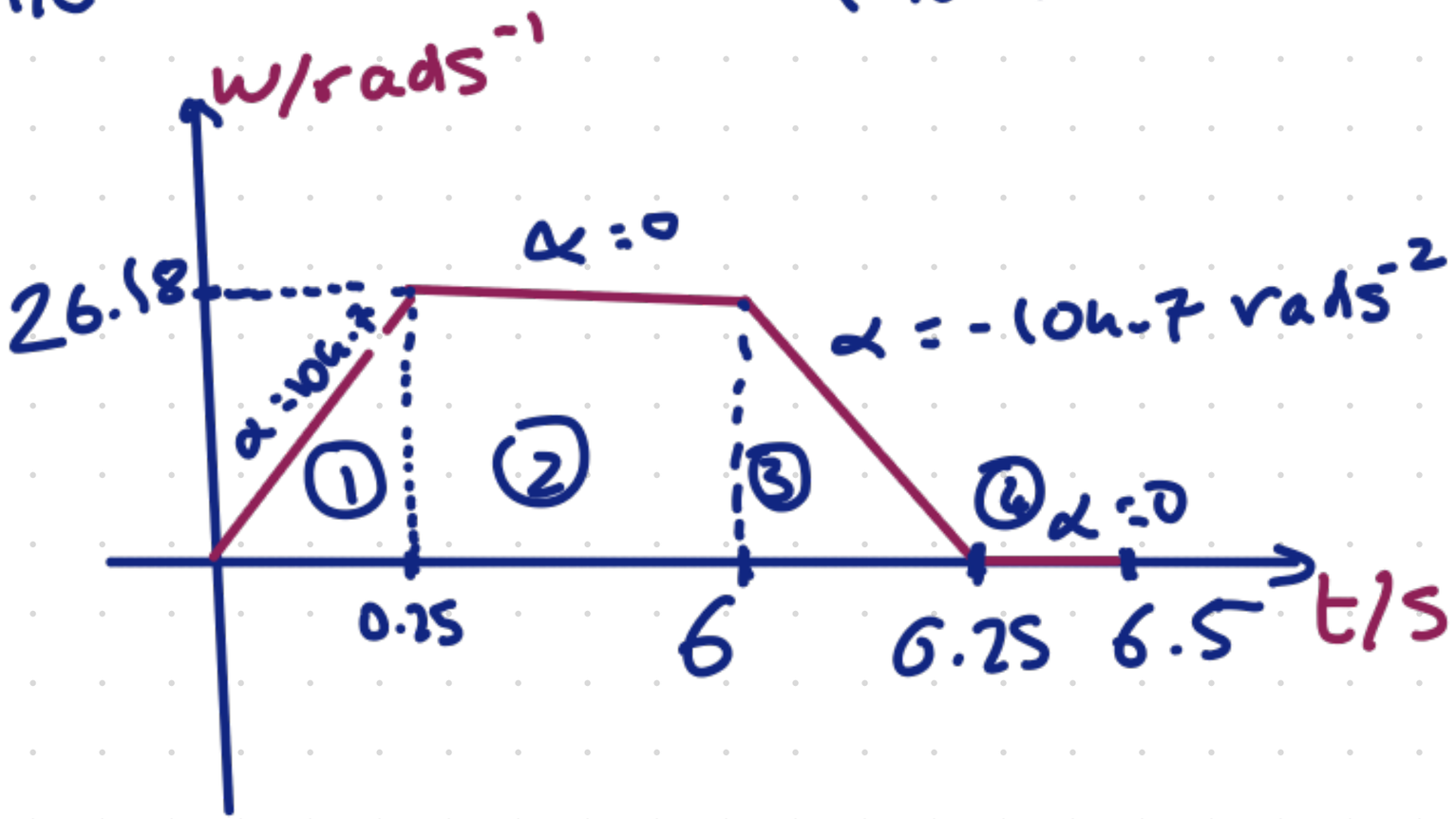
$$J_L = \left(\frac{1}{2} m_{screw} r_{screw}^2\right) + \left(m_{load}\left(\frac{P}{2\pi}\right)^2\right)$$

$$= 6.4\times10^{-7} + 0.4\times\left(\frac{0.008}{2\pi}\right)^2 = 1.29\times10^{-6}\ Nm^2 = \underline{12.9\ Ncm^2}$$

$$J_{tot} = 12.9\times2 = 25.8\ Ncm^2$$

$$T_L = \frac{mgP}{2\pi\eta} = \frac{0.4 \times 9.81 \times 0.008}{2\pi \times 0.5} = \underline{1 \text{ Ncm}}$$

Velocity Profile: (tot dist = 20 cm)



$$T_m = T_L + J_{tot}\,\alpha = 1 + 25.8\times10^{-5}\,\alpha$$

$$T_1 = 1 + 25.8\times10^{-5}(10\pi.7) = 1.027 \quad N\,cm$$

$$T_2 = 1 \quad N\,cm$$

$$T_3 = 1 + 25.8\times10^{-5}(-10\pi.7) = 0.973 \quad N\,cm$$

$$T_h = 1 \quad N\,cm$$

$$T_{rms} = \sqrt{\frac{1.027^2\times0.25 + 5.75 + 0.973^2\times0.25 + 0.25}{6.5}}$$

$T_{rms} \approx 1\,Ncm \Rightarrow$ 25GA-370 motor with encoder

End effector

Suction Cup diameter = 25 mm $\Rightarrow$ Area = $\frac{\pi}{4} 25^2 = 490\ mm^2$

$= 0.00049\ m^2$

$F = mg = PA$; $50 \times 10^{-3} \times 9.81 = P(0.00049)$

$\Rightarrow P = 1\ kPa$ $\Rightarrow$ 12V vacuum pump 66 kPa

feeding

$$\Sigma T = J\alpha$$

$$T_m - T_L = (J_m + J_L)\,\alpha$$

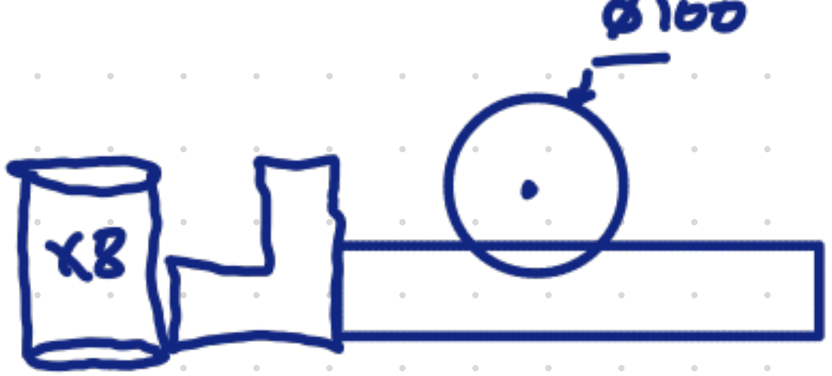
$$T_m = T_L + (J_m + J_L)\alpha$$

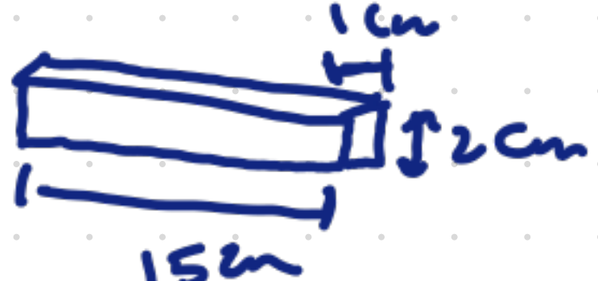
$$J_L = \tfrac{1}{2}\left(700 \times \pi (0.05)^2 \times 0.01\right)(0.05)^2$$

$$+ \left(700 \times 0.15 \times 0.02 \times 0.01\right)(0.05)^2$$

$$+ 8 \times 0.05 \times 0.05^2 = 1.12 \times 10^{-3}\ kgm^2$$

$$J_m = J_L \quad ; \underline{J_{tot} = 2.24 \times 10^{-3}\ kgm^2}$$

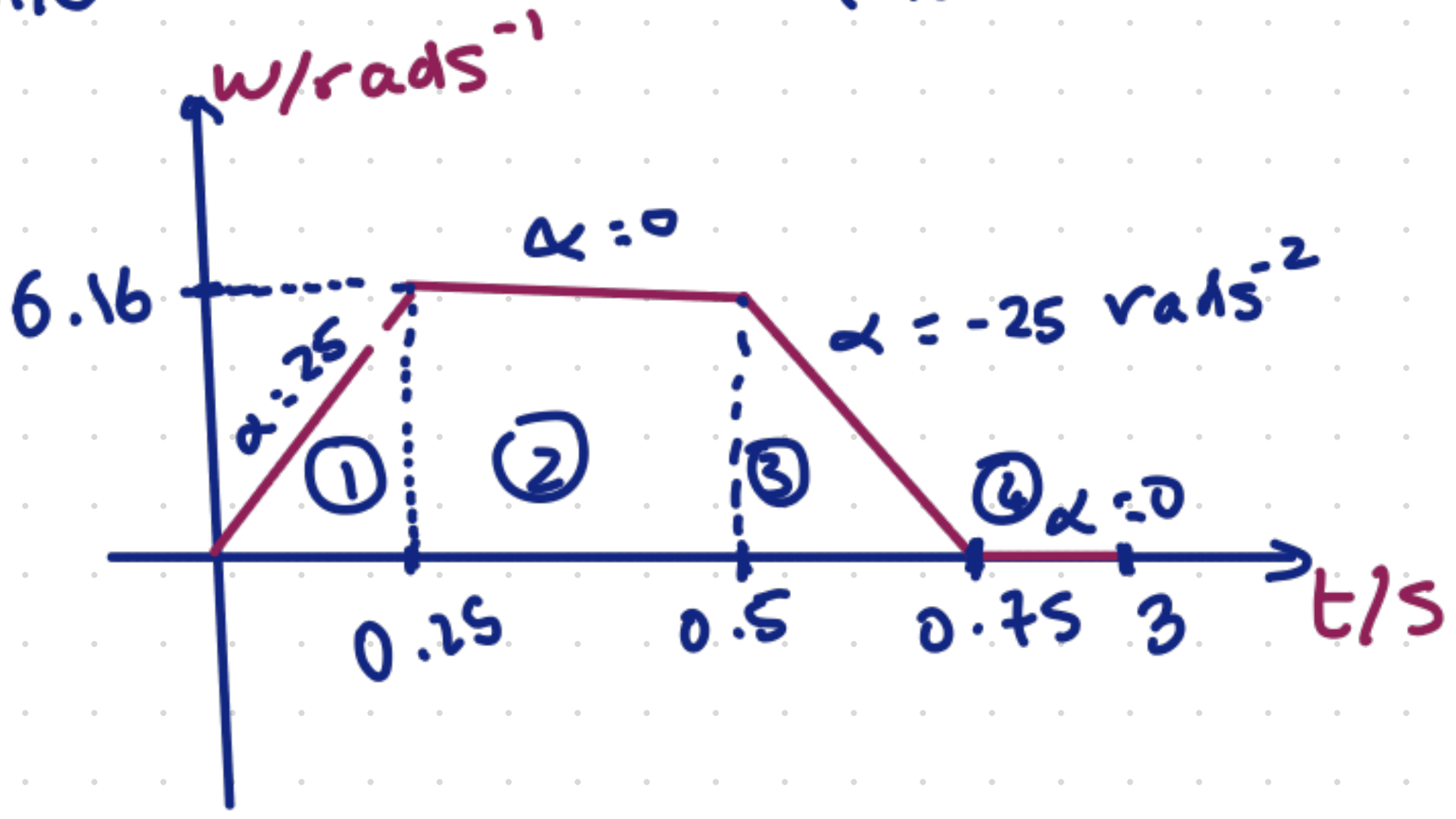






$T_L = mg\mu R \approx 8 \times 0.05 \times 9.81 \times 0.5 \times 0.05 = \underline{0.0981 \ Nm}$

Velocity Profile : (tot dist ≈ 10 cm)



$$T_m = 0.0981 + 2.24 \times 10^{-3} \alpha$$

$$T_1 = 0.0981 + 2.24 \times 10^{-3} (25) = 0.1541 \ Nm$$

$$T_2 = 0.0981 \ Nm$$

$$T_3 = 0.0981 + 2.24 \times 10^{-3} (-25) = 0.0421 \ Nm$$

$$T_4 = 0.0981 \ Nm$$

$$\boxed{T_{rms} = 1.027 \ kgFcm} \Rightarrow MG996 \ Servo$$